NIKOLA MILOSAVLJEVIC

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TECH STACK

Languages: Rust, Golang, Python, C# (.NET)
Databases: SQL (PostgreSQL, Mssql), NoSQL (MongoDB, Neo4j)
Automation: GitHub Actions, Ansible, Bazel
IaaC (Infra-as-code): Terraform, Pulumi
Cloud: AWS, GCP, Azure, Kubernetes, Docker

WORK EXPERIENCE

DFINITY Stiftung **Zurich, Switzerland**
Site Reliability Engineer / Software Engineer ***Sep 2022 – Present***

- **Designed and implemented a scalable service discovery module** powering the company's observability platform, synchronizing global system state, integrating with **Prometheus** and **VictoriaMetrics**, and generating dynamic observability configurations.
- **Designed, built, and operated the company's end-to-end logging infrastructure**; led migration from **AWS** to on-prem **Kubernetes**, reducing costs by **~$100K/month** and supporting **~10M prod logs/hour and ~20M dev logs/hour**.
- Designed and implemented features **reducing operational burden for node providers**, including widely used direct node swaps and migrations that **reduced key rotation time from days to seconds while improving security and flexibility**.
- Built a linear optimization tool for decentralized network design, **enabling constraint-based topology generation**, **trade-off modeling**, and **capacity planning for target network states**.
- **Developed core components of the company's cloud platform**, including a **Web2 control panel** for provisioning and managing sovereign cloud environments and a **Web3 on-chain module** for billing and **payments with stablecoin and asset swap integration**.
- Built a **local testing driver** using QEMU and libvirt **to spin up production-like environments**, enabling **1:1 testing** against production and **shortening feedback loops** while **improving code quality**.
- Developed an **automated release qualification pipeline**, executing pre/post-upgrade **performance tests** in production-like environments to **quantify regressions and guide release decisions**.
- Contributed to **developer experience** and **internal tooling** at DFINITY, **managing testnet infrastructure**, **improving build systems**, **building on-call utilities**, and **streamlining release** and feature rollouts.

Sotex Solutions (outsourced to Schneider Electric Sevilla) **Novi Sad, Serbia**
Fullstack Software Engineer ***May 2021 – Sep 2022***

- Built a **billing module** for municipalities to generate utility bills across Kuwait, Sevilla, and the Netherlands, **integrating IoT smart meters** and **handling terabytes of data** with **reliable, large-scale processing**.
- Developed smart meter **orchestration systems** enabling users and technicians to **remotely configure, monitor and diagnose devices** using **industry-standard protocols**, **improving observability** and **operation control at scale**.
- Refactored **project delivery workflows** by **containerizing applications** and **implementing Infrastructure-as-code** for automated environment provisioning, improving deployment consistency, scalability and release speed.

Sotex Box **Novi Sad, Serbia**
Founding Engineer ***May 2024 – Sep 2025***

- Built and scaled a **low-bandwidth ad delivery platform** for constrained networks, running on minimal infrastructure (**~$80/month**) while supporting **~10K concurrent devices** and **generating ~$15K/month** in revenue.

EDUCATION

Faculty of Technical Sciences **Novi Sad, Serbia**
Bachelor's and Master's Degrees ***Sept 2018 – Dec 2023***

- **Bachelor's** GPA: 9.49/10 - graduated in the top 5%
- **Master's** GPA: 9.75/10 - graduated in the top 1%
- Worked as a teaching assistant during the last year of Bachelor's and Master's. Was involved with cyber/security and microservice architecture.